Particle Physics:
A Very Short Introduction

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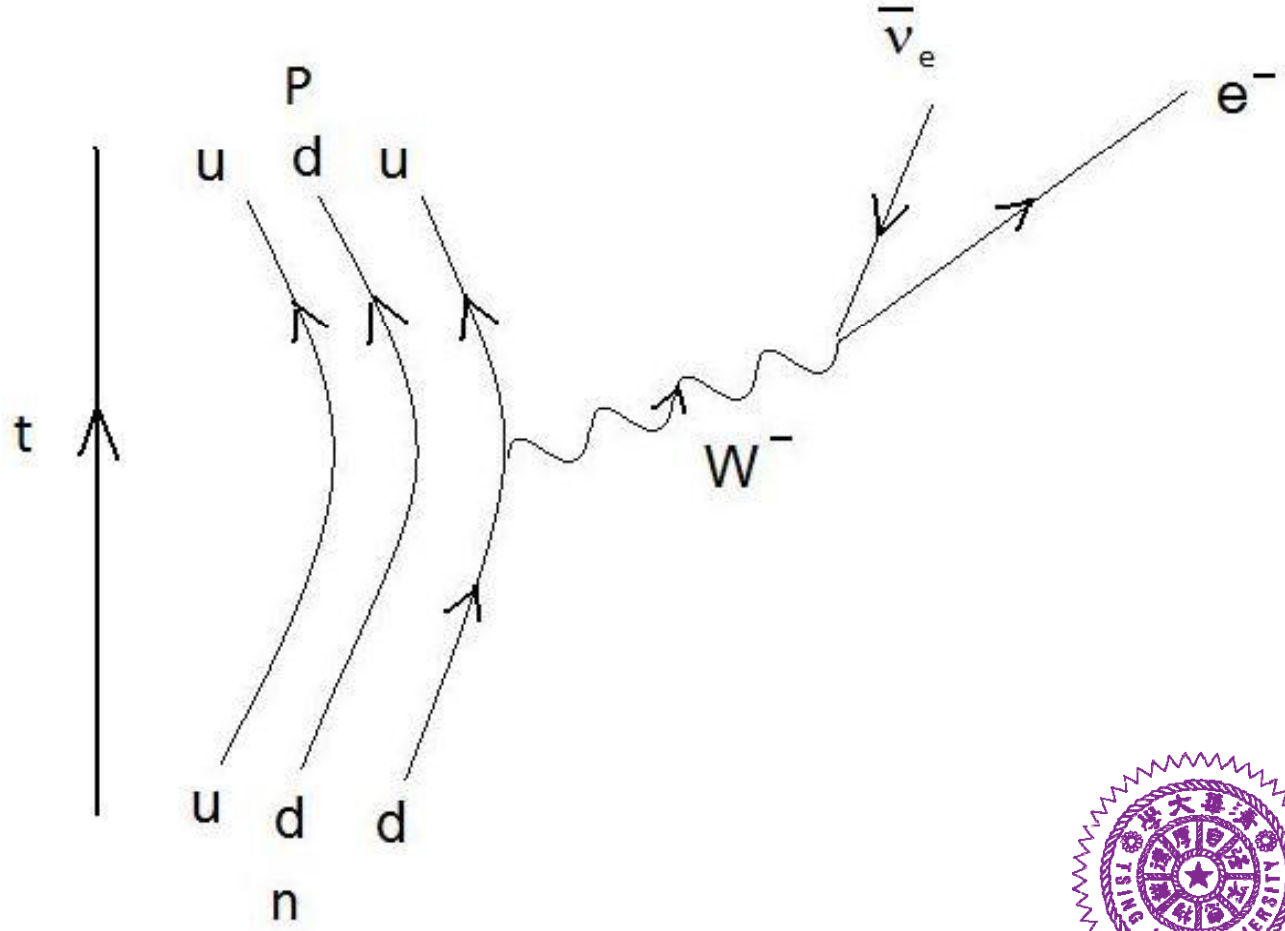
Outline

- Fundamental Force
- Standard Model
- God Particle
- The Frontier

Fundamental Force

- Feynman Diagram: β decay

$$n \rightarrow p + e^- + \bar{\nu}_e$$



Fundamental Force

- Four fundamental force in nature
  - Gravitation → General Relativity
    - Strength of force: 10^{-42}

  - Weak interaction → GWS Theory(Electroweak Theory)
    - Strength of force: 10^{-13}
    - Exchanging $W^{\pm}$ and Z^0 bosons
  - Electromagnetic interaction → Quantum Electrodynamics(QED)
    - Strength of force : 10^{-2}
    - Exchanging photon
  - Strong interaction → Quantum Chromodynamics(QCD)
    - Strength of force : 10
    - Exchanging gluon

Fundamental Force

- Gravitation is smaller than other interactions, we do not discuss it in particle physics

- The most two important theories in modern physics
  - Relativity
  - Quantum mechanics

- Special Relativity + Quantum Mechanics
  → Quantum Field Theory(QFT)
  - QFT is a tool to study fundamental force
  - Each particle can be associated with a field

Standard Model

Standard Model of

FUNDAMENTAL PARTICLES AND INTERACTIONS

The Standard Model summarizes the current knowledge in Particle Physics. It is the quantum theory that includes the theory of strong interactions (quantum chromodynamics or QCD) and the unified theory of weak and electromagnetic interactions (electroweak). Gravity is included on this chart because it is one of the fundamental interactions even though not part of the "Standard Model."

FERMIONS

matter constituents
spin = 1/2, 3/2, 5/2, ...

Leptons spin = 1/2

Flavor	Mass GeV/c^2	Electric charge
ν_e electron neutrino	$<1\times10^{-8}$	0
e electron	0.000511	–1
ν_μ muon neutrino	<0.0002	0
μ muon	0.106	–1
ν_τ tau neutrino	<0.02	0
τ tau	1.7771	–1

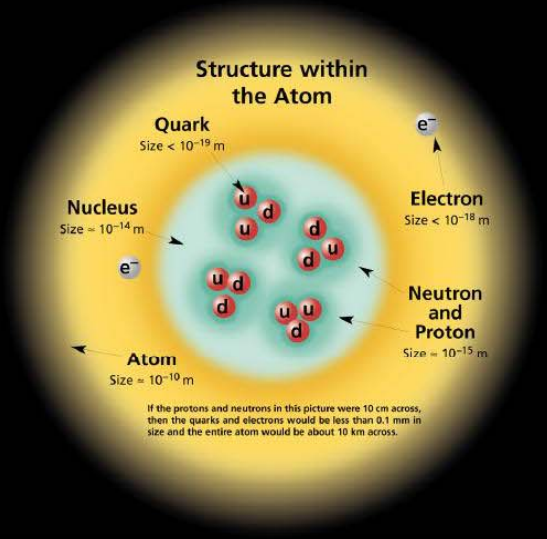
Quarks spin = 1/2

Flavor	Approx. Mass GeV/c^2	Electric charge
u up	0.003	2/3
d down	0.006	–1/3
c charm	1.3	2/3
s strange	0.1	–1/3
t top	175	2/3
b bottom	4.3	–1/3

Spin is the intrinsic angular momentum of particles. Spin is given in units of $\hbar$, which is the quantum unit of angular momentum, where $\hbar = h/2\pi = 6.58\times10^{-25}$ GeV s $= 1.05\times10^{-34}$ J s.

Electric charges are given in units of the proton's charge. In SI units the electric charge of the proton is 1.60×10^{-19} coulombs.

The **energy** unit of particle physics is the electronvolt (eV), the energy gained by one electron in crossing a potential difference of one volt. **Masses** are given in GeV/c^2 (remember $E = mc^2$), where 1 GeV = 10^9 eV = 1.60×10^{-10} joule. The mass of the proton is 0.938 GeV/c^2 = 1.67×10^{-27} kg.



BOSONS

force carriers
spin = 0, 1, 2, ...

Unified Electroweak spin = 1

Name	Mass GeV/c^2	Electric charge
γ photon	0	0
W^-	80.4	–1
W^+	80.4	+1
Z^0	91.187	0

Strong (color) spin = 1

Name	Mass GeV/c^2	Electric charge
g gluon	0	0

Color Charge
Each quark carries one of three types of "strong charge," also called "color charge." These charges have nothing to do with the colors of visible light. There are eight possible types of color charge for gluons. Just as electrically-charged particles interact by exchanging photons, in strong interactions color-charged particles interact by exchanging gluons. Leptons, photons, and ***W*** and ***Z*** bosons have no strong interactions and hence no color charge.

Quarks Confined in Mesons and Baryons
One cannot isolate quarks and gluons; they are confined in color-neutral particles called **hadrons**. This confinement (binding) results from multiple exchanges of gluons among the color-charged constituents. As color-charged particles (quarks and gluons) move apart, the energy in the color-force field between them increases. This energy eventually is converted into additional quark-antiquark pairs (see figure below). The quarks and antiquarks then combine into hadrons; these are the particles seen to emerge. Two types of hadrons have been observed in nature: **mesons** $q\bar{q}$ and **baryons** qqq.

Residual Strong Interaction
The strong binding of color-neutral protons and neutrons to form nuclei is due to residual strong interactions between their color-charged constituents. It is similar to the residual electrical interaction that binds electrically neutral atoms to form molecules. It can also be viewed as the exchange of mesons between the hadrons.

PROPERTIES OF THE INTERACTIONS

Property \ Interaction	Gravitational	Weak (Electroweak)	Electromagnetic (Electroweak)	Strong: Fundamental	Strong: Residual
Acts on:	Mass – Energy	Flavor	Electric Charge	Color Charge	See Residual Strong Interaction Note
Particles experiencing:	All	Quarks, Leptons	Electrically charged	Quarks, Gluons	Hadrons
Particles mediating:	Graviton (not yet observed)	W^+ W^- Z^0	γ	Gluons	Mesons
Strength relative to electromag for two u quarks at: 10^{-18} m	10^{-41}	0.8	1	25	Not applicable to quarks
Strength relative to electromag for two u quarks at: 3×10^{-17} m	10^{-41}	10^{-4}	1	60	
for two protons in nucleus	10^{-36}	10^{-7}	1	Not applicable to hadrons	20

Baryons qqq and Antibaryons $\bar{q}\bar{q}\bar{q}$
Baryons are fermionic hadrons.
There are about 120 types of baryons.

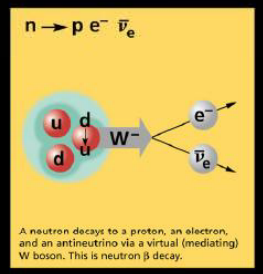
Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
p	proton	uud	1	0.938	1/2
$\bar{p}$	anti-proton	$\bar{u}\bar{u}\bar{d}$	–1	0.938	1/2
n	neutron	udd	0	0.940	1/2
Λ	lambda	uds	0	1.116	1/2
Ω^-	omega	sss	–1	1.672	3/2

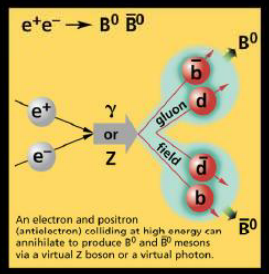
Mesons $q\bar{q}$
Mesons are bosonic hadrons.
There are about 140 types of mesons.

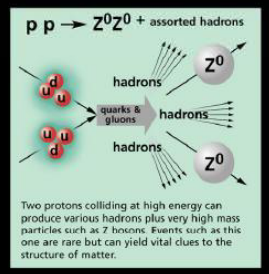
Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
π^+	pion	$u\bar{d}$	+1	0.140	0
K^-	kaon	$s\bar{u}$	–1	0.494	0
ρ^+	rho	$u\bar{d}$	+1	0.770	1
B^0	B-zero	$d\bar{b}$	0	5.279	0
η_c	eta-c	$c\bar{c}$	0	2.980	0

Matter and Antimatter
For every particle type there is a corresponding antiparticle type, denoted by a bar over the particle symbol (unless + or – charge is shown). Particle and antiparticle have identical mass and spin but opposite charges. Some electrically neutral bosons (e.g., Z^0, γ, and $\eta_c = c\bar{c}$, but not $K^0 = d\bar{s}$) are their own antiparticles.

Figures
These diagrams are an artist's conception of physical processes. They are ***not*** exact and have ***no*** meaningful scale. Green shaded areas represent the cloud of gluons or the gluon field, and red lines the quark paths.







The Particle Adventure
Visit the award-winning web feature *The Particle Adventure* at **http://ParticleAdventure.org**

This chart has been made possible by the generous support of:
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Standard Model

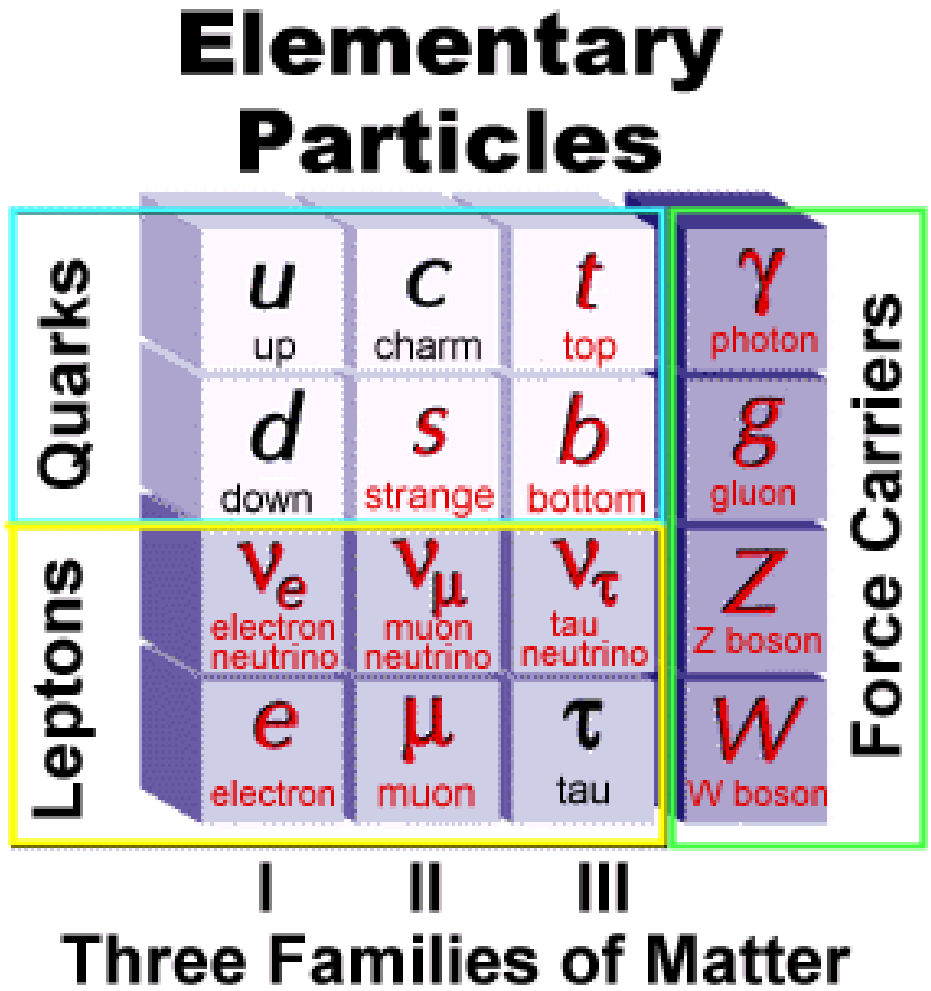
- Fundamental particles are consist of half-integer spin fermions

- Two types
  - Fermions
    - Half-integer spin
    - Anti-symmetric
    - Obey Pauli exclusion principle
    - Fermi-Dirac statistics

  - Bosons
    - Integer spin
    - Symmetric
    - Bose-Einstein statistics

Standard Model

- Quarks(three generations)
  - First: up(u), down(d)
  - Second: charm(c), strange(s)
  - Third: top(t), bottom(b)
- Leptons(three generations)
  - First: electron(e^-), electron-neutrino(n_e)
  - Second: muon(μ^-), muon-neutrino (n_μ)
  - Third: tauon(τ^-), tauon-neutrino(ν_τ)
- Gauge Bosons(mediator)
  - Strong interaction: gluon(g)
  - Weak interaction: $W^\pm$ and Z^0 bosons
  - Eletromagnetic interaction: photon(γ)
  - Gravitation: graviton



Copy from SLAC website

- Higgs Boson(?)

Standard Model

- Some properties of quarks
  - Flavor –Isospin(I), it is similar to spin
    - Eg. Nucleon(from Heisenberg)

$$I = \frac{1}{2}$$

$$I_3 = \frac{1}{2}, \text{isospin up} \rightarrow p = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

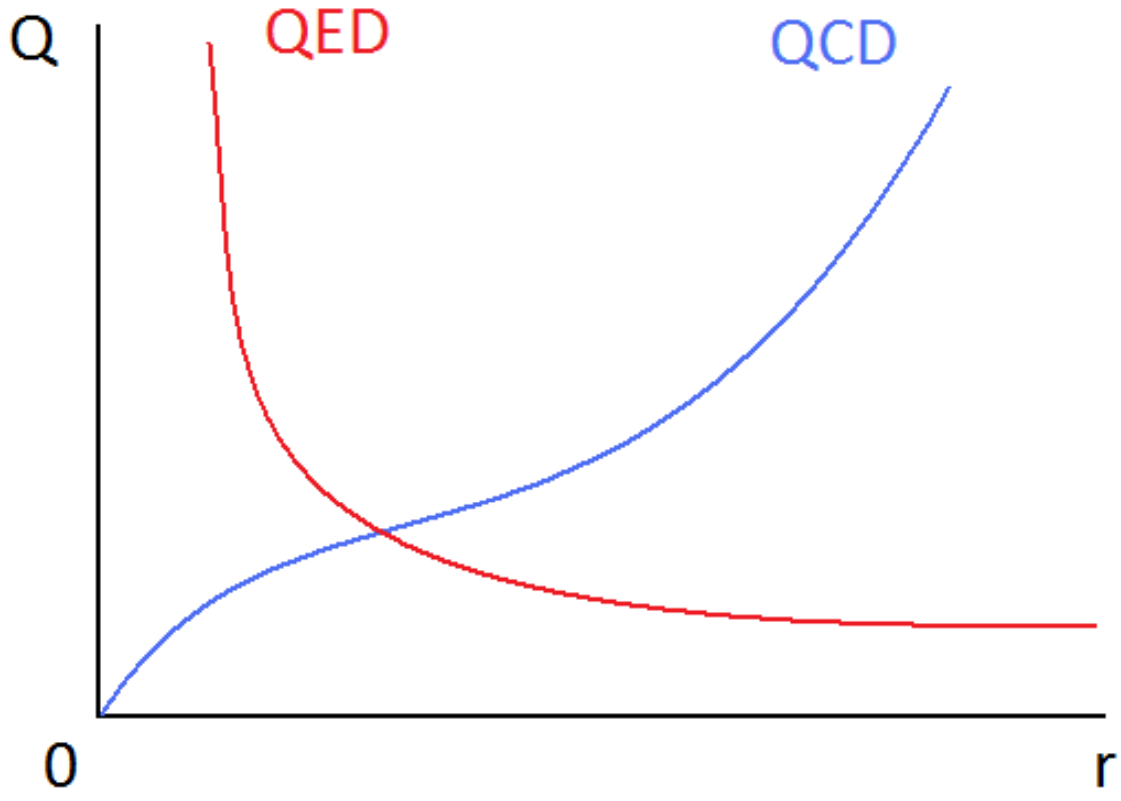
$$I_3 = -\frac{1}{2}, \text{ isospin down} \rightarrow n = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

  - Spin(S)

  - Color
    - Red(r)
    - Green(g)
    - Blue(b)

Standard Model

- Asymptotic freedom of quarks
  - It is unlike Coulomb's interaction in QED



Standard Model

- Hadron
  - Baryon(qqq)
  - Meson($q\bar{q}$)
- Antimatter and Antiparticles
  - From special relativity

$$E^2 = p^2c^2 + m^2c^4$$

$$\Rightarrow E = \pm\sqrt{p^2c^2 + m^2c^4}$$

    - + sign: particle
    - - sign: antiparticle

SM Predicts the God Particle

- Statue of Zeus

Copy from the internet

God Particle

- Higgs boson is god particle which is not observed in experiment yet

- Spontaneous symmetry breaking in GWS theory

Copy from Wikipedia website

- Higgs particle is the last particle which is not observed yet in SM

- Higgs particle gives mass for everything
  - Higgs mechanisim gives mass to vector bosons(spin-1 bosons like: $W^{\pm}$ and Z^0), quarks and leptons
  - In GWS theory, fermions have no mass
    By spontaneous symmetry breaking, higgs particle would gives mass to fermions

The Frontier

The Frontier

- GUT(Grand Unification Theory)

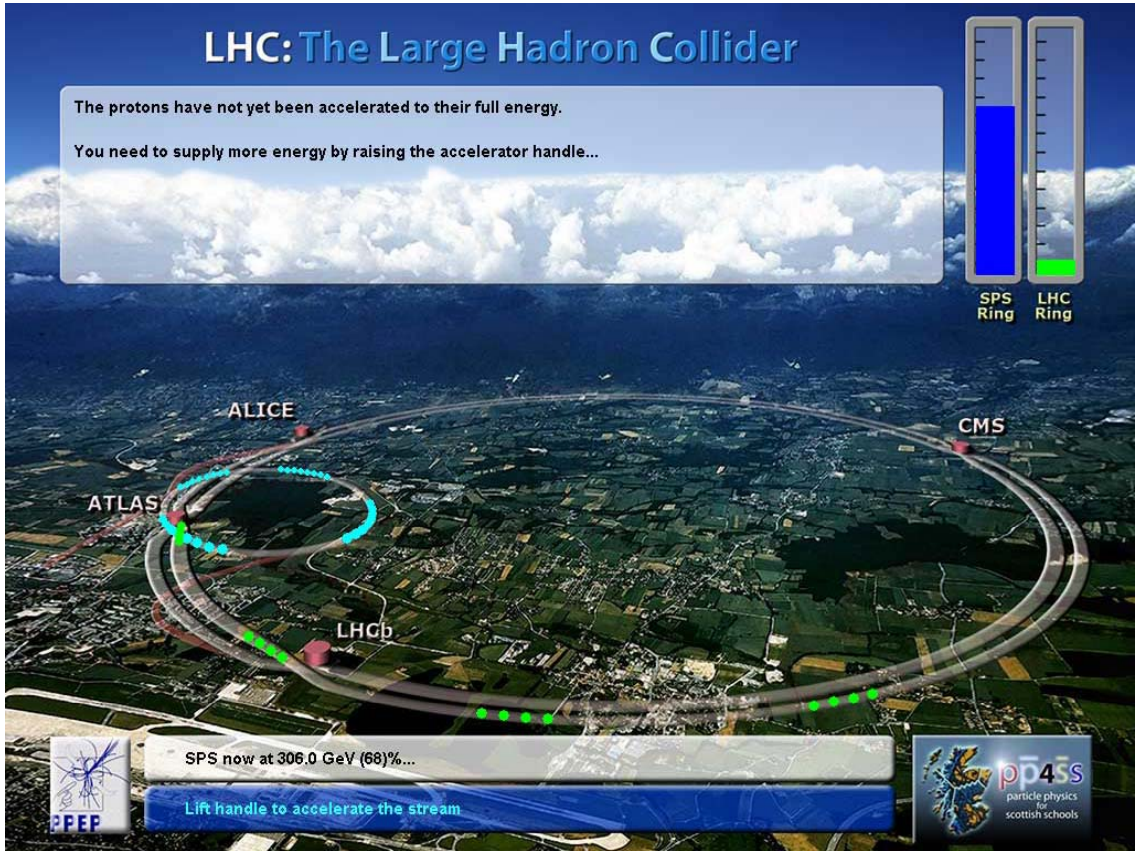
- SUSY(Supersymmetry)
  - Fermions <—> Bosons(fermion boson symmetry)

- TOE(Theory Of Everything)
  - => String Theory(26-dim) + SUSY → Superstring Theory(11-dim)
  - => 11-dim M-Theory(predicted by Superstring Theory)

- Particle accelerator is the useful device to use
  - It used electric field to propel the charged particles to high speed
  - The simple form is CRT TV
  - Basic types
    - Linear accelerator
    - Circular accelerator

The Frontier

- Collider
  - Using two particle beams flying from opposite directions and the particles would collide.

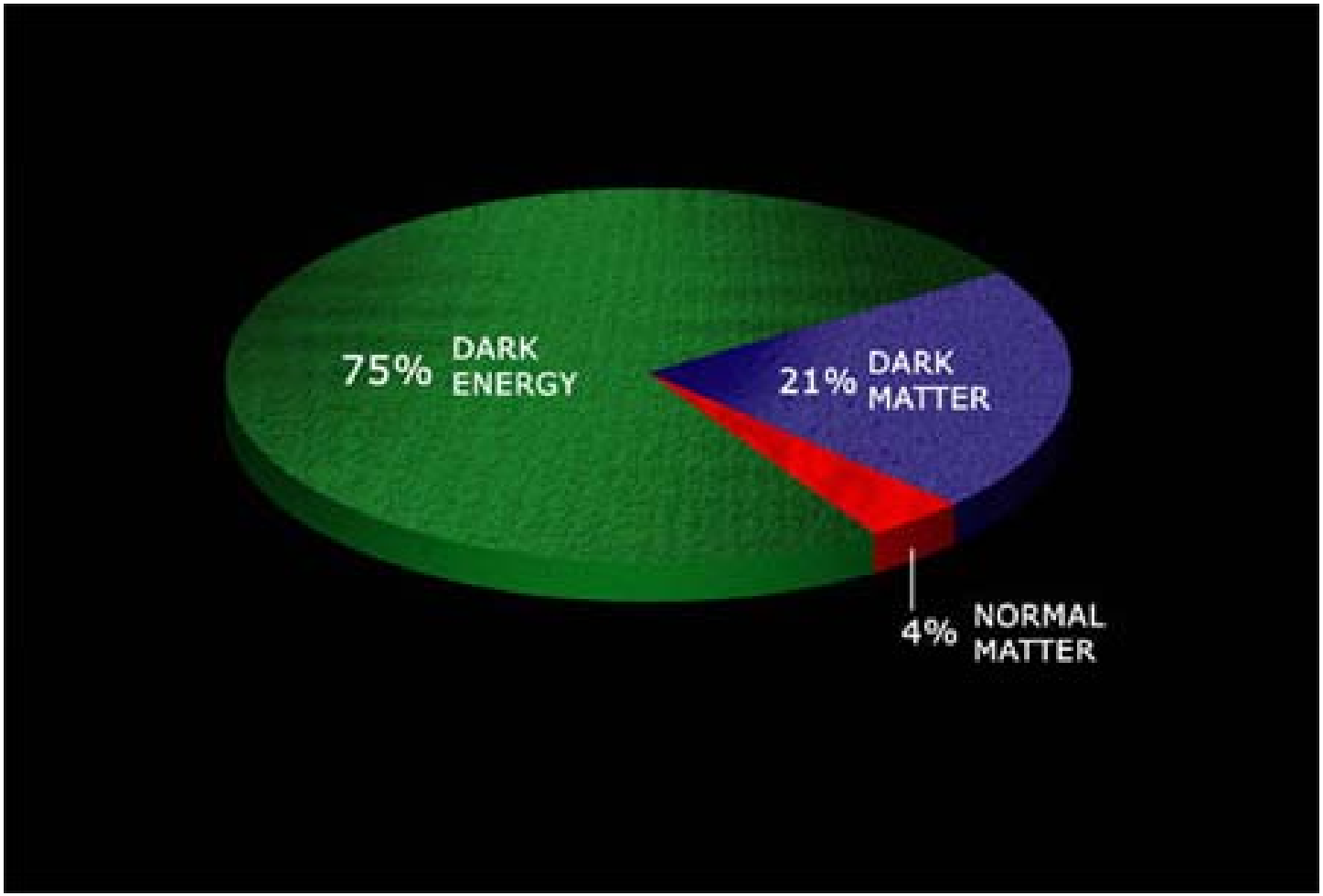


Copy from the internet

- LHC
  - Large Hadron Collider(LHC) located @ CERN, Geneva
  - It will turn on in this month
  - Purpose
    - To find the Higgs particles
    - To find the supersymmetric particles
    - Why the magnitude of gravity is so weak
    - Why matter and antimatter are not symmetric
    - Etc.

The Frontier

- Dark Matter
  - The particle we do not know yet
- Dark Energy
  - All that we can not explain



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